



Seminar

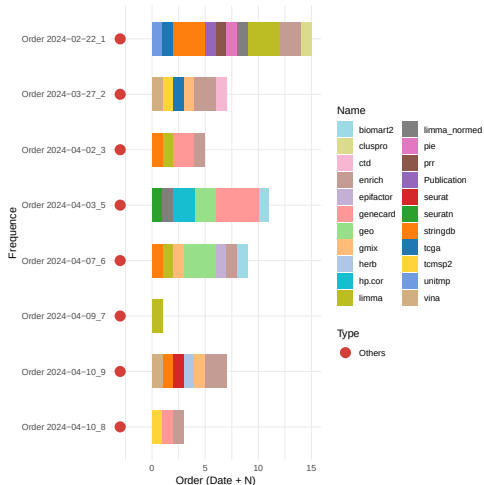
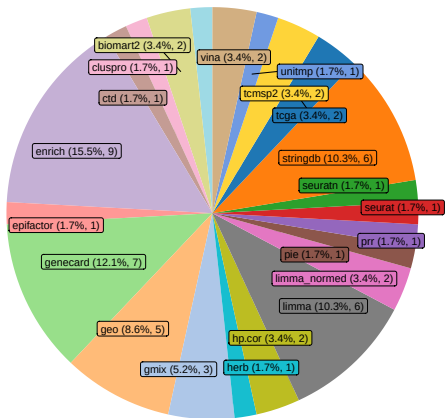
Lichuang Huang

Wie-Biotech

2024-05-15

Tasks summary

Overview 2024-04



New Methods

- batman, BATMAN-TCM
- tcmsp2, TCMSP
- dl, D-GCAN (python), drug-likeness
- ctd, Comparative Toxicogenomics Database
- cluspro, ClusPro, protein-protein docking

Article sharing

scFEA

Method

A graph neural network model to estimate cell-wise metabolic flux using single-cell RNA-seq data

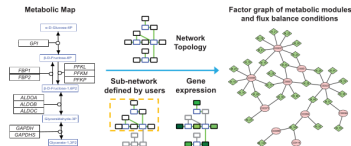
Norah Alghamdi,^{1,6} Wennan Chang,^{1,2,6} Pengtao Dang,^{1,2} Xiaoyu Lu,¹ Changlin Wan,^{1,2} Silpa Gampala,³ Zhi Huang,^{1,2} Jiashi Wang,¹ Qin Ma,⁴ Yong Zang,^{1,5} Melissa Fishel,³ Sha Cao,^{1,5} and Chi Zhang^{1,2}

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Overview

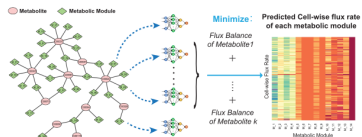
- 构建数据: Metabolic modules
- 通量估计: Cell-wise Flux Rate
- 下游分析: Others

A Network reduction and reconstruction into factor graph

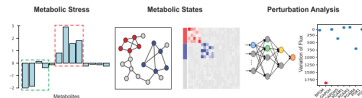


B Flux Estimation

$$L = \sum_{\text{Cells } i} \sum_{\text{Metabolites } j} \text{Flux Balance Loss} + \sum_{\text{Modules } k} \text{Non-Negative Loss} + \sum_{\text{Modules } l} \text{Inconsistency with Gene Expression} + \text{Flux Scale}$$



C Downstream Analysis



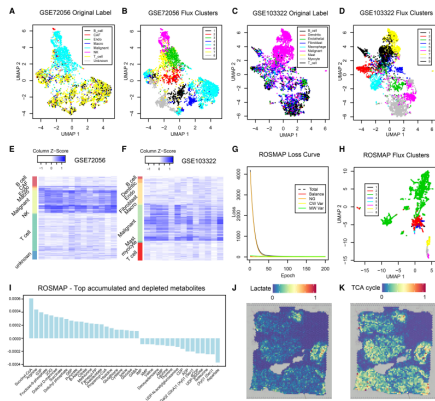
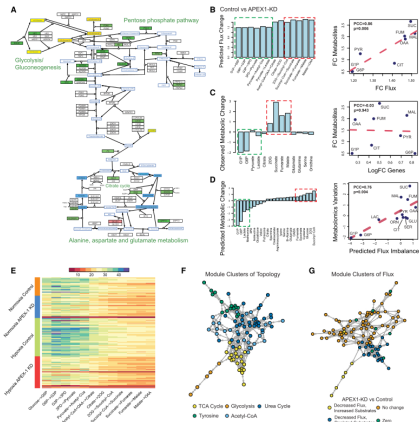
Situation

自发性脑出血（ICH）后调控小胶质细胞代谢的关键基因 XXX



- 单细胞数据
- 代谢水平
- 代谢物和基因的关系
- 但没有空间代谢组数据

Examples



Integrate with Seurat

After scFEA done, we can read results into Seurat again to perform downstream analysis.

```
##R
predFlux <- read.csv('./output/Seurat_gluta_flux.csv', header = T, row.names = 1)
predFlux <- data.matrix(predFlux)
predFlux0 <- t(predFlux)
```

```
# add flux as a new assay
obj[["FLUX"]] <- CreateAssayObject(counts = predFlux0)
```

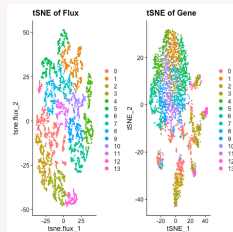
```
R[write to console]: Warning:
R[write to console]: Feature names cannot have underscores ('_'), replacing with dashes ('-')
```

Change default assay to FLUX to perform cell clustering based on metabolic flux and plot tSNE.

```
##R
DefaultAssay(obj) <- 'FLUX'
obj <- FindVariableFeatures(obj, selection.method = "vst", nfeatures = 2000, verbose = F)
obj <- ScaleData(obj, features = rownames(obj), assay = 'FLUX', verbose = F)
obj <- RunPCA(obj, features = VariableFeatures(object = obj), npcs = 10, reduction.name = 'pca.flux', v
obj <- FindNeighbors(obj, dims = 1:2, verbose = F)
obj <- FindClusters(obj, resolution = 0.5, verbose = F)
obj <- RunTSNE(obj, dims = 1:2, assay = 'FLUX', reduction.name = "tsne.flux", verbose = F)
DimPlot(obj, reduction = "tsne.flux") + ggtitle('tSNE of Flux')
```

tSNE of Flux

```
##R
library(patchwork)
p1 <- DimPlot(obj, reduction = "tsne") + ggtitle('tSNE of Gene')
p2 <- DimPlot(obj, reduction = "tsne.flux") + ggtitle('tSNE of Flux')
p2 + p1
```



Schedule

Schedule

- Finish orders
- New methods

Thanks